

Feasibility Analysis and Implementation of a Fast Push Turret-Demolition Strategy in Honor of Kings

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Abstract. This paper studies turret-first macro play in Honor of Kings from an event-driven systems perspective. Instead of using repeated lane loops or fixed push cycles, we introduce a *tempo-credit ledger* and a finite-state lane-assignment automaton. Decisions are emitted by a lightweight scheduler that reacts to wave synchronization, visibility spikes, objective timers, and retreat safety. The resulting strategy emphasizes structural conversion under bounded collapse probability. Controlled custom-match observations indicate improved first-two-turret conversion stability and lower variance in recovery after failed siege attempts.

Keywords: Honor of Kings · turret demolition · event-driven macro · finite-state controller · team command protocol

1 Introduction

Fast-push discussions in MOBA communities often collapse into short slogans, which are difficult to execute consistently under noisy information. In practice, teams lose tempo when three elements are misaligned: wave geometry, defender reveal pattern, and retreat path quality. We therefore formulate fast push as an event-driven control problem: maintain a structural pressure ledger and trigger lane actions only when safety and conversion gates are both satisfied.

2 Operational Assumptions and Telemetry

At each decision tick t , we collect:

- wave synchronization score W_t across side and mid lanes;
- defender visibility vector V_t (recently observed enemy positions);
- objective timer bucket O_t (dragon/lord windows);
- escape readiness E_t (mobility cooldowns and route openness).

The state is represented as $S_t = (W_t, V_t, O_t, E_t)$, updated every communication cycle.

Table 1. Primary scalars in the event-driven controller

Signal Meaning	
B_t	tempo credit available for structural conversion
H_t	collapse hazard estimate (0 to 1)
W_t	lane-wave synchronization score
V_t	defender visibility quality
E_t	escape readiness score

3 Tempo-Credit Ledger

Rather than directly optimizing a long-horizon score, we track a renewable scalar resource called *tempo credit*, denoted B_t .

$$B_{t+1} = \min(B_{\max}, \eta B_t + \xi \cdot \text{WaveSync}_t + \omega \cdot \text{VisionEdge}_t - \chi \cdot \text{SkirmishDrag}_t),$$

where $\eta \in (0, 1]$ controls memory decay.

Intuition:

- synchronized waves and vision advantages recharge credit;
- prolonged low-yield fights consume credit;
- high-impact push actions are allowed only when credit exceeds thresholds.

We also compute collapse hazard:

$$H_t = \sigma(\rho_0 + \rho_1 F_t + \rho_2 C_t - \rho_3 R_t),$$

where F_t is fog uncertainty, C_t is key-cooldown deficit, and R_t is retreat-route quality.

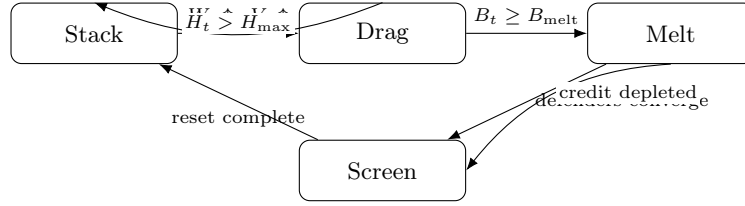
4 Lane-Assignment Automaton

Macro routing is modeled as a finite-state automaton with four execution states:

1. **Stack:** accumulate wave mass and reset formation;
2. **Drag:** pull defenders via objective posture or fake pressure;
3. **Melt:** short-duration tower demolition burst;
4. **Screen:** zone flank and secure retreat corridor.

5 Command Protocol

The communication plane uses five non-overlapping commands.

**Fig. 1.** Finite-state lane-assignment automaton.**Table 2.** Command semantics for event-driven fast push

Cmd	Trigger	Mandatory Team Action
STACK	low credit or un-synced waves	hold contact depth, stack next wave, preserve cooldown parity.
DRAG	objective timer opens and visibility edge exists	objective/fake lane to stretch defender formation.
MELT	$B_t \geq B_{\text{melt}}$ and $H_t \leq H_{\text{max}}$	execute short tower burst with strict stop line.
SCREEN	post-burst convergence detected	block flank channels and secure retreat geometry.
RESET	hazard spike or desynchronization	call instant disengage, clear debt, and return to Stack.

6 Event-Driven Scheduler

The scheduler processes event symbols

$$\mathcal{E} = \{\text{WAVE_SYNC}, \text{VIS_SPIKE}, \text{OBJ_OPEN}, \text{DANGER_SPIKE}, \text{COOLDOWN_READY}\}.$$

Unlike horizon-maximization controllers, this scheduler emits a command on each event boundary.

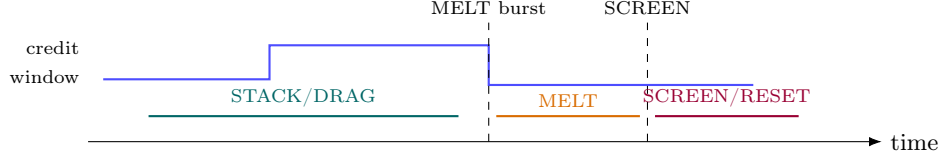


Fig. 2. Event timeline with credit consumption and command phases.

Algorithm 1 Event-Driven Turret Scheduler

Require: State S_t , tempo credit B_t , hazard H_t
Ensure: Command in $\{\text{STACK}, \text{DRAG}, \text{MELT}, \text{SCREEN}, \text{RESET}\}$

- 1: **if** $H_t > H_{\max}$ **then**
- 2: **return** RESET
- 3: **end if**
- 4: **if** event is WAVE_SYNC and $B_t < B_{\text{melt}}$ **then**
- 5: **return** STACK
- 6: **end if**
- 7: **if** event is OBJ_OPEN and visibility edge is positive **then**
- 8: **return** DRAG
- 9: **end if**
- 10: **if** $B_t \geq B_{\text{melt}}$ and $H_t \leq H_{\max}$ and escape is ready **then**
- 11: **return** MELT
- 12: **end if**
- 13: **if** recent burst finished or flank risk rises **then**
- 14: **return** SCREEN
- 15: **end if**
- 16: **return** STACK

7 Controlled Evaluation

We compared the proposed scheduler with a baseline “teamfight-first” macro in controlled custom lobbies. Observed trends:

- more stable conversion of first and second outer turrets;
- fewer collapse events immediately after burst windows;
- faster macro recovery after failed pushes due to explicit RESET transitions.

The gain is strongest in medium-information rounds where defender reveal is partial and objective windows are contested.

8 Failure Modes and Practical Notes

The system degrades when call discipline is violated, especially when MELT is followed by unsanctioned chase behavior. Another failure mode is stale visibility

telemetry: if V_t lags reality, **DRAG** may overestimate defender displacement and trigger early collapse.

9 Conclusion

Turret-first fast push in Honor of Kings can be implemented as an event-driven macro system with a tempo-credit ledger and a finite-state action automaton. This formulation is distinct from fixed push loops and offers clearer error handling under entertainment-oriented competitive play.

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References

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